SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIE
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: Theory of Computation COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Deign Finite Automata DFA, NFA, ε-NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:50

Design Task

Code of Conduct:

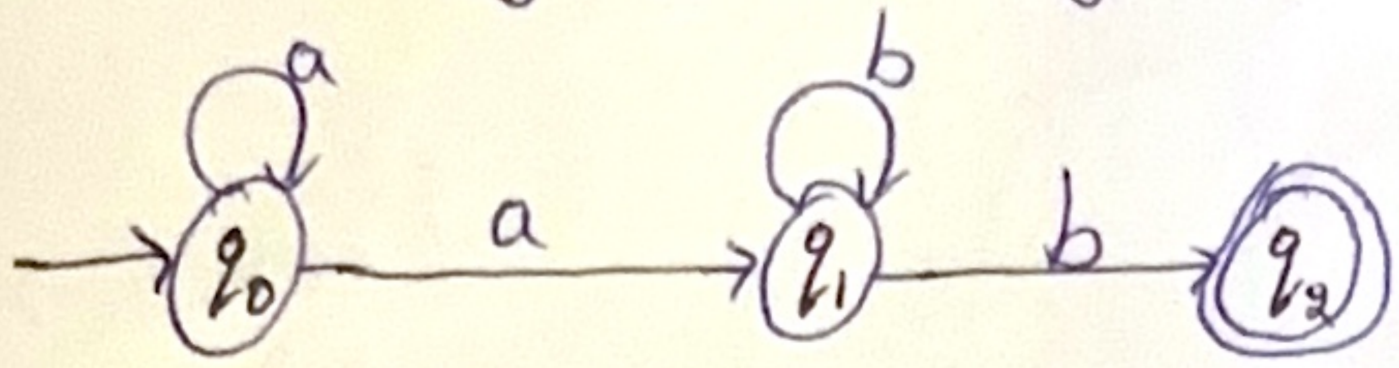
I Y. Yoganarasimha 192311254 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Regular Expression.

Given:-

To design the regular expression for the language represented by the given automate.



* From q_0 we can have any number of 'a' s including zero and then one 'a'. to go to q_1

* From q_1 we can have any number of 'b's including zero and then one 'b' to reach final state q_2.

∴ Regular expression = $a^* a b^* b$.

2. Consider the following grammer.

$S \rightarrow aSc \,/\, J \,/\, u \,/\, \varepsilon$

$T \rightarrow bTc \,/\, \varepsilon$

$u \rightarrow aUb \,/\, \varepsilon$

A. Language defined by the grammer:-

From $T -$ $bTC/\varepsilon \Rightarrow T$ generate b^*c^*

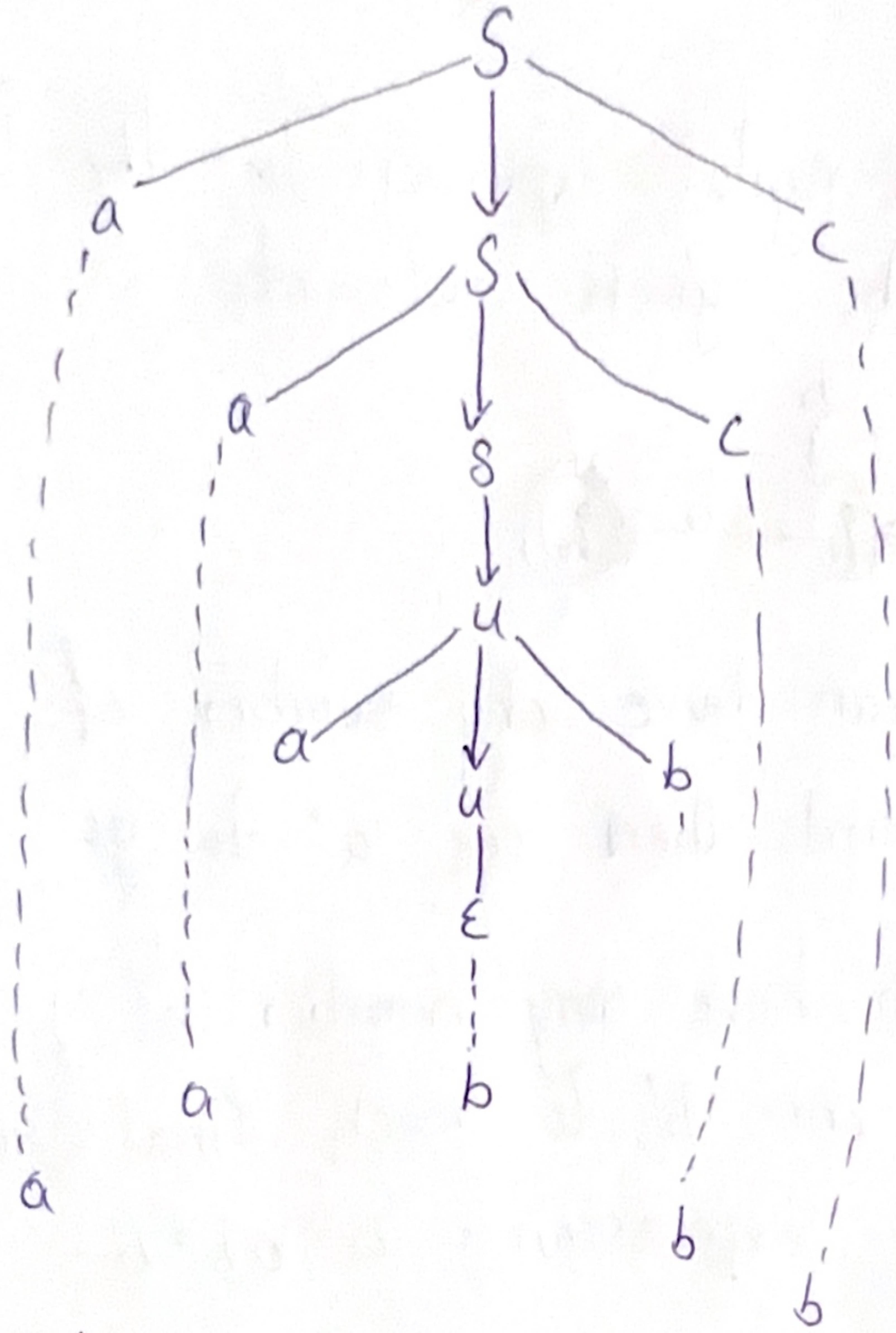
From $u \rightarrow aUb/\varepsilon \Rightarrow u$ generate a^*b^*

From $S \rightarrow aSc/T/u/\varepsilon$ -

$\Rightarrow S$ generates $(asc)^* (J/H/\varepsilon)$

$\Rightarrow L(S) = (asc)^* (b^*c^*/a^*b^*/\varepsilon)$

B. Derivation tree for the word aaa bbc.



Leaves read left to right gives aaa bb

ⓒ Ambiguity:-

In $S \to aSc$, each 'a' must be matched with exactly one 'c'. Hence the number of a's and c's must be equal and the pairing its unique.

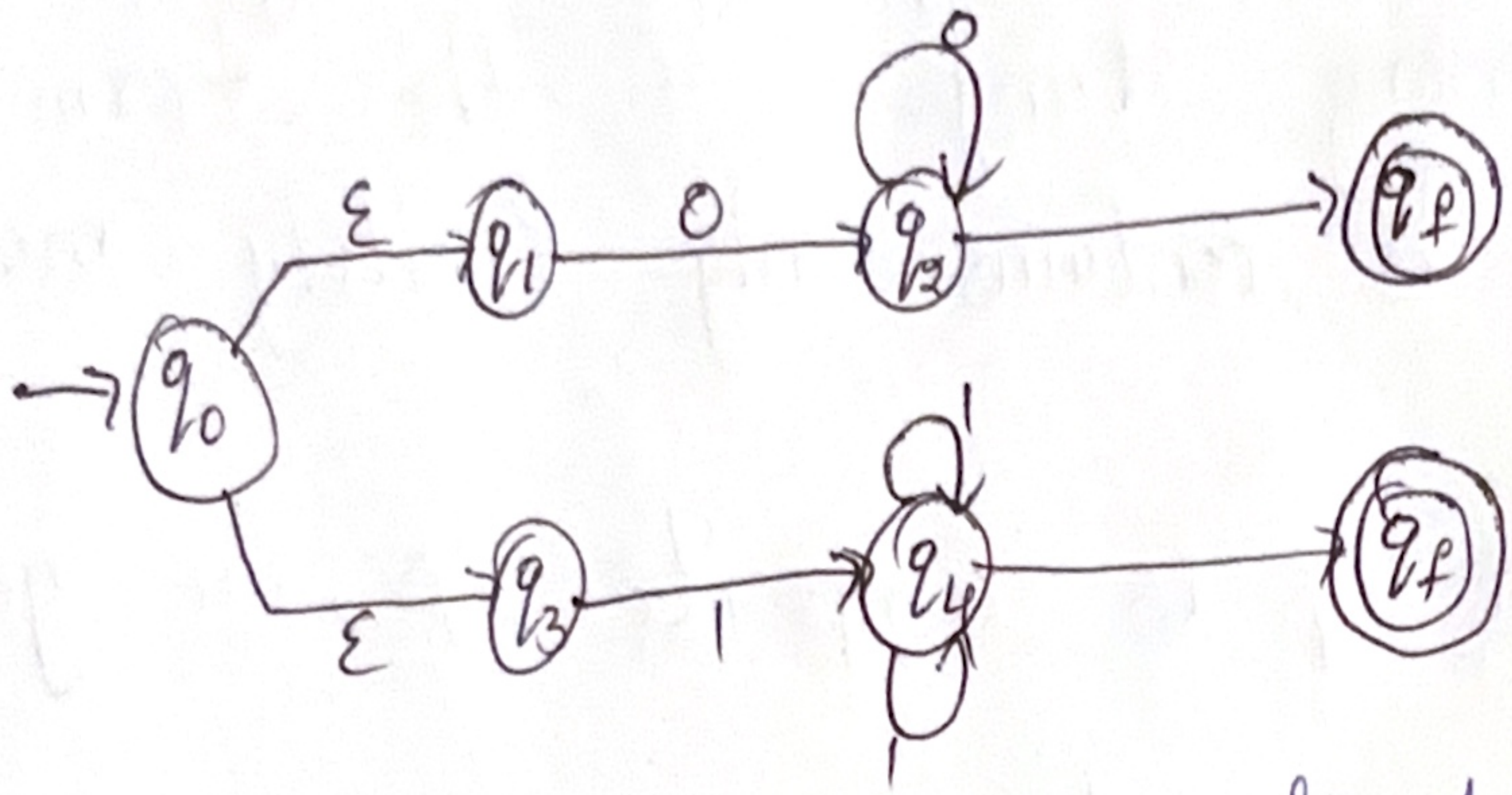
Also the choice between T or u is deterministic for a given string. Therefore the grammer is not ambiguous.

3. Non-deterministic finite automate with ε-transitions to represent the language denoted by the regular Expression.

$$00^*1 + 11^*0$$

Let $R = 00^*1 + 11^*0$

This is a union of 2 expressions, we design ε-NFA for both and combine using ε-moves.



q_0 is start state, q_f is final state. This ε-NFA accepts strings of the form 00^*1 or 11^*0.

(4) Show that if L_1 & L_2 are regular then $(L_1 + L_2)$ also regular.

Sol:- Let L_1 & L_2 be regular languages. Then there exist DFAs $M_1 = (Q, \Sigma, \delta, q_{1}, F_1)$ for L_1 and $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ for L_2. Construct a new DFA $M = (Q, \Sigma, \delta, q_0, F)$ as follow.

$Q = Q_1 * Q_2$.

$q_0 = (q_1, q_2)$

$F = (F_1 \times Q_2) \cup (Q_1 \times F_2)$

$\delta((p, r), a) = (\delta(p, a), \delta_2(r, a))$ for all $p \in Q_1$. $r \in Q_2$, $a \in \Sigma$.

M accepts a string w if either M_1 accepts w or M_2 accepts w.

Hence $L(M) = L_1 + L_2$. Thus $(L_1 + L_2)$ is regular.

II) Is the language $\{a^p / p \text{ is prime}\}$ regular? Justi[fy] state

Sol:- Let $L = \{a^p / p \text{ is prime}\}$

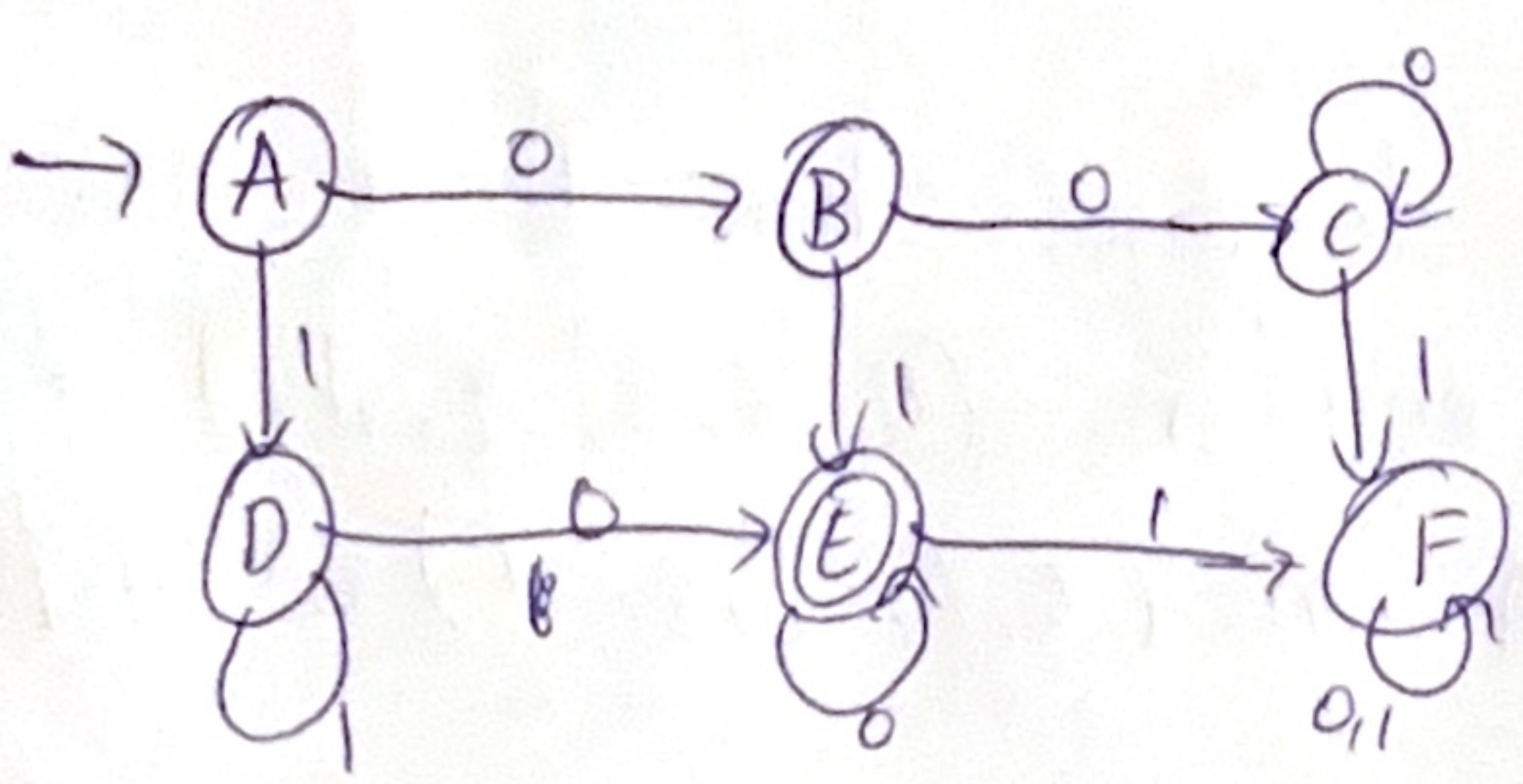
By pumping lemma, any regular language must have infinitely many strings of the form a^n for $n \geq p$. Here L contains only finitely many such strings.

Choosing a^q where q is composite $\geq p$, we get contradiction.

$\therefore L$ is not regular.

5. Design minimized automata.

Given DFA.



Transition table of minimized DFA.

state	Input 0	Input 1
→A	B	D
B	C	E
C	C	F
D	E	D
*E	E	E
F	F	F

* Final state.

Sl:- Take Filling method.

Mark pairs.

(C,F) : distinguishable (C on 0→C, F on 0→F)

(A,D) : distinguishable (A on 0→B, D on 0→E (final))

(B,E) : distinguishable (B on 0→C, E on 0→E)

(A,B) : distinguishable (A on 1→D, B on 1→E)

(B,F) : distinguishable (B on 0→C, F on 0→F)

(D,F) : distinguishable (D on 0→E, F on 0→F)

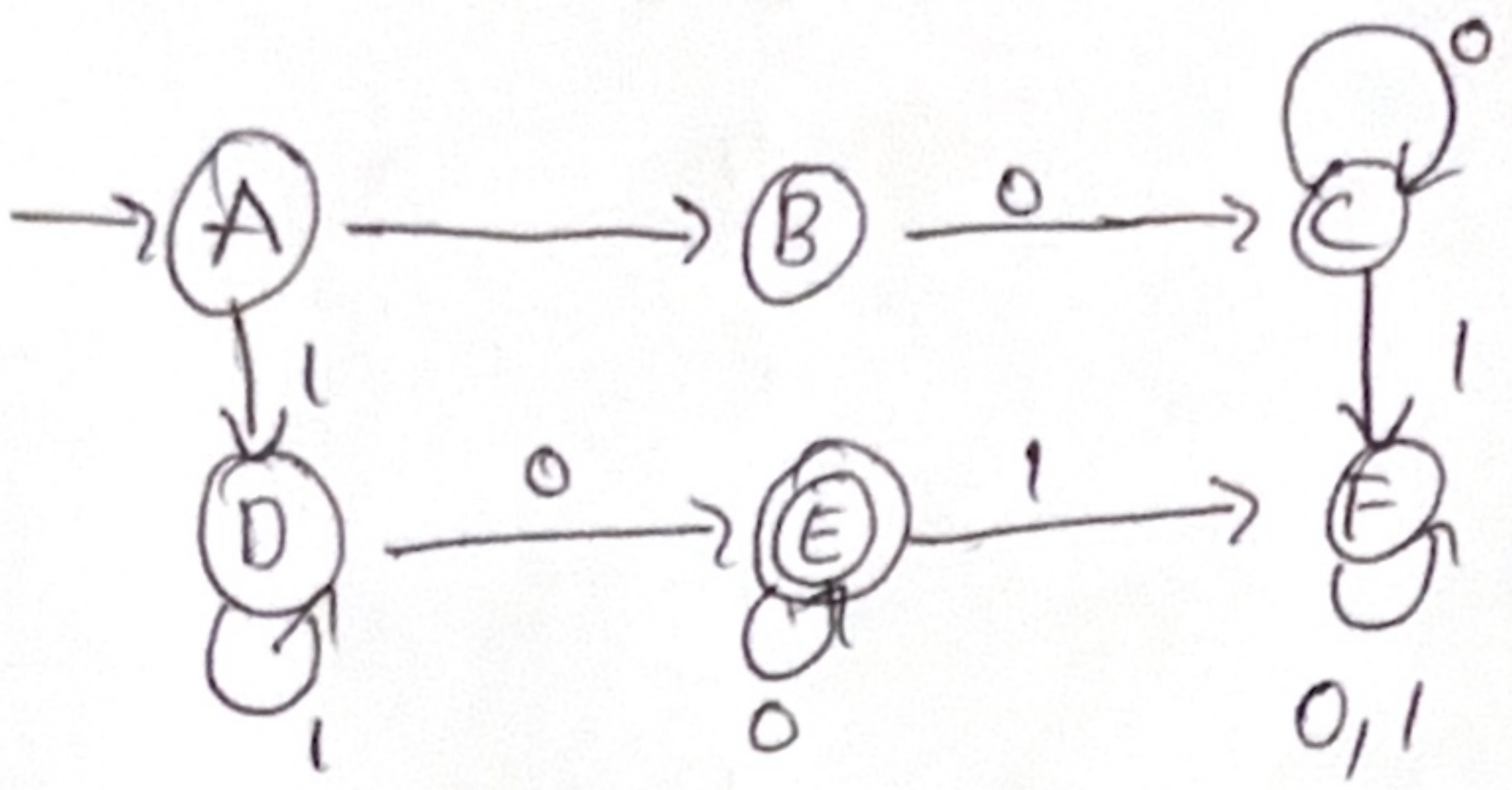
(A,C) : distinguishable (A on 1→D, C on 1→F)

(D,E) : distinguishable (D on 0→E, E on 0→E) but D on 1→D, E on 1→F).

∴ No 2 states are equivalent.

All states are pairwise distinguishable.

Minimized DFA.



∵ No 2 states are equivalent, the minimum DFA has 6 states.

start state :- A

Final state :- E

— x —